Noise and Outliers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Noise and Outliers

> Prepare Noise and Outliers: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Calculate **mean** and **median**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Overfitting Intuition

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Training and Testing

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

Concept to explain

Mistakes and **unusual value**s should be **investigate**d, not automatically erased.

Term	Meaning
noise	Variation or interference that obscures the information relevant to a task.
outlier	An observation far from a typical pattern under a stated comparison; it is not automatically an error.
duplicate	A repeated record or copy; copies are not independent new observations.
investigate	Collect and examine evidence to answer a question rather than assuming a cause.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 17 as a two-session investigation using fictional data. Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects.

Keep all values while investigating whether 18 is a genuine long activity or a recording error. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

9.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can calculate **mean** and **median** and **investigate** an **unusual value** before changing it.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Noise and Outliers using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Most books: 40–300 pages; one record: 3,200 pages.

2. Check the source: extra zero, box set, or real long work?

3. Repair confirmed error; keep real rare case and document it.

4. Retest to learn whether the system handles legitimate extremes.

Recalculate, retrace, or restate the reasoning in your own representation:

> CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

> ### The teaching case
>
> Recorded lengths are 90,95,100,105,900 pages.

> Is 900 automatically wrong?

Exact explanation to model

> ### Why the result follows
>
> No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

Explain the added worked example

Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects.

Keep all values while investigating whether 18 is a genuine long activity or a recording error.

The script prints 6.0 and 3. sorted returns a sorted list without changing the original. This **median** expression works for a nonempty odd-length list only. For four sorted values, average the second and third.

Do not use this odd-length shortcut for every dataset.

Key terms

Term	Meaning
Mean	Sum of numeric values divided by their count
Median	The middle of sorted values or mean of the two middle values
Unusual value	A value far from most others in a dataset
Variation	Differences among recorded values

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Replace 18 with 8. Calculate the new mean and median.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Noise and Outliers

A fictional library records five book-**process**ing times in minutes.

The facts we will use

Supplied facts

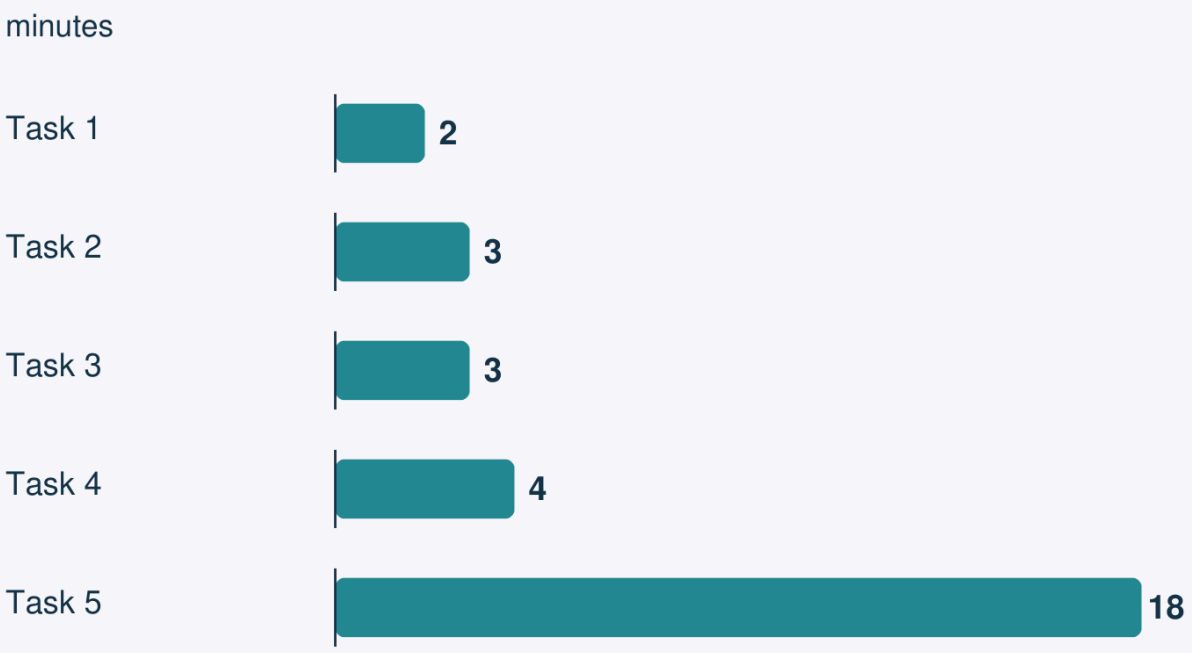
Sorted times: 2,3,3,4,18.

Supplied facts

All are recorded measurements; no **evidence** yet shows 18 is a mistake.

AI ACADEMY / AI-2/17

Retain unusual recorded measurements



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/17

Noise and Outliers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Mean

Sum 30; divide by 5: mean 6 minutes.

2. Median

The middle of five sorted values is 3 minutes.

3. Investigate 18

Check the source record and whether an unusually slow task occurred.

4. Document repair

Change a value only if evidence supports a correction; preserve the raw record.

CHANGE ONE THING

An authorised log confirms 18 was a transcription error for 3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Mean	Sum 30; divide by 5: mean 6 minutes.
2. Median	The middle of five sorted values is 3 minutes.
3. Investigate 18	Check the source record and whether an unusually slow task occurred.
4. Document repair	Change a value only if evidence supports a correction; preserve the raw record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

What this example does not establish

Keep the stated limit

Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

Change one thing in the pictured case

Change one thing

Change: An authorised log confirms 18 was a transcription error for 3.

Predicted result

Prediction: Corrected **mean** is 3 minutes and **median** is 3.

Why the result follows

Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Ask while showing the picture

Ask while showing the picture · 1

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Noise and Outliers, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Recorded lengths are 90,95,100,105,900 pages.

2. State the decision to explain

Write the question to explain: Is 900 automatically wrong?

3. Reveal the worked reasoning

Reveal after **prediction**s: No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

Hold back the independent task

Keep for an independent attempt: Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to **investigate**.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Sum 20; **mean** 4; **median** remains 3. Exit: No. Zero would be a measured summary; no numeric **observation**s are available.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Find **median**s of [9, 1, 3] and [8, 2, 4, 6].

Reasoning to listen for

Teacher check: Sorted first list 1, 3, 9 has **median** 3. Sorted second 2, 4, 6, 8 has **median** (4 + 6)/2 = 5.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Mean median** and **unusual value**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Removing 18 merely because it raises the **mean** would hide **evidence**. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding **variation**.

Pictorial practice question

> If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

Reasoned teacher answer

Median 3 describes the middle; also report the unusual 18-minute task and the **mean** 6 rather than concealing spread.

Changed-case reasoning: Corrected **mean** is 3 minutes and **median** is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Noise and Outliers investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Audit four suspicious records using keep / correct / remove / ask labels. Require a reason and a source check for every action.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: Removing 18 merely because it raises the **mean** would hide **evidence**. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: 2, 4, 4, 5, 15 has sum 30, **mean** 6 and **median** 4. Replacing 15 with 5 gives sum 20, **mean** 4 and **median** 4.

A convenient shortcut preserves **mean**ing: Taking the middle item of an unsorted list can give the wrong **median**. For [18, 2, 4, 3, 3], the middle item is 4, but the sorted **median** is 3. Sort first.

For an empty list, there is no **mean** or **median** under this lesson's rules.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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Assess and explain the answers

> Use the Noise and Outliers questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

> **Teacher key · Worked case answer**
>
> No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

Independent question

> **Independent question · withhold the key**
>
> Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to **investigate**.

Expected independent answer

> **Teacher key · Expected independent answer**
>
> 900 may be a genuine long book or a recording error. Check the original record and task scope before changing it; unusual does not automatically **mean** wrong.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Every **outlier** is an error.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Check edition, catalogue and source entry. Retain 960 if verified; correct only if **evidence** establishes an error, recording original value, new value and reason.

Automatic deletion can remove a legitimate category and distort the intended collection.

Explanations for the additional workbook examples

W1 Compute a mean

Workbook question · W1

Find the **mean** of 4, 4, 6, 6 and 10.

Teacher reasoning · W1

Sum 30 divided by 5 gives 6.

Teacher reasoning · W1

Add all five **observation**s: 4 + 4 + 6 + 6 + 10 = 30. Divide by the number of **observation**s, five, to obtain a **mean** of 6. Keep the count beside the calculation.

Do not divide by the three distinct values; repeated **observation**s still count.

W2 Compute two medians

Workbook question · W2

Find **median**s of [9, 1, 3] and [8, 2, 4, 6].

Teacher reasoning · W2

Sorted first list 1, 3, 9 has **median** 3. Sorted second 2, 4, 6, 8 has **median** (4 + 6)/2 = 5.

W3 Repair an explanation

Workbook question · W3

A learner deletes 18 because it is above the **mean**. What **evidence** is needed before correcting or excluding it?

Teacher reasoning · W3

Check original measurement and collection rules. Being above the **mean** alone is not an error. Retain the raw record and document any justified correction.

W4 Compare summaries independently

Workbook question · W4

For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

Teacher reasoning · W4

Original sum 20, **mean** 4, **median** 2. Revised sum 10, **mean** 2, **median** 2. The large value changes **mean** more than **median** in this example.

W5 Report missing evidence

Workbook question · W5

A dataset is empty. Is reporting a **mean** of 0 justified?

Teacher reasoning · W5

No. Zero would be a measured summary; no numeric **observation**s are available.

Assess independent transfer

Independent question · withhold the key

Selected case: For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

Selected case answer

Teacher key · Selected case answer

Original sum 20, **mean** 4, **median** 2. Revised sum 10, **mean** 2, **median** 2. The large value changes **mean** more than **median** in this example.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Noise and Outliers to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Keep legitimate unusual books so the recommender can serve unusual readers.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 18: Missing Data. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Order number cards physically. Extension: write a **median** function with explicit empty, odd and even cases and test each.

Independent W4 **evidence**: Original sum 20, **mean** 4, **median** 2. Revised sum 10, **mean** 2, **median** 2. The large value changes **mean** more than **median** in this example. Record whether the learner needed no help, a hint or explanation.

Fresh retry: 2, 4, 4, 5, 15 has sum 30, **mean** 6 and **median** 4. Replacing 15 with 5 gives sum 20, **mean** 4 and **median** 4.

Next week: Graphs and honest scales. Reuse this week's explicit contracts and **evidence**.

Plan the delayed return

Suggested return points in this level: ai-2/18, ai-2/20, ai-2/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Noise and Outliers lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Calculate **mean** and **median**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Find **median**s of [9, 1, 3] and [8, 2, 4, 6].

Supported practice answer

Reasoned practice answer: Sorted first list 1, 3, 9 has **median** 3. Sorted second 2, 4, 6, 8 has **median** (4 + 6)/2 = 5.

Check every learner before discussion: Use the worked case for **Mean median** and **unusual value**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Removing 18 merely because it raises the **mean** would hide **evidence**. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: An authorised log confirms 18 was a transcription error for 3. Require a **prediction** and a reason before comparing. Expected result: Corrected **mean** is 3 minutes and **median** is 3.

Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

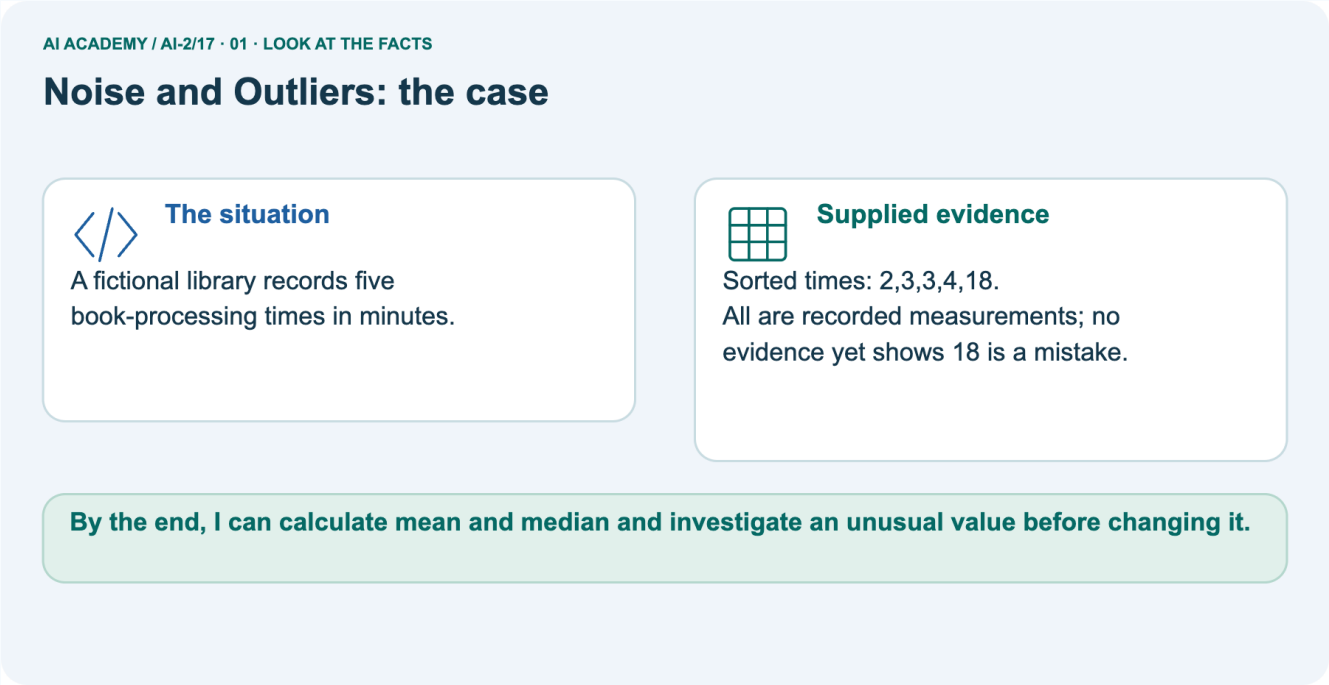
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Visual concept journey

> Show one Noise and Outliers graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Noise and Outliers: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

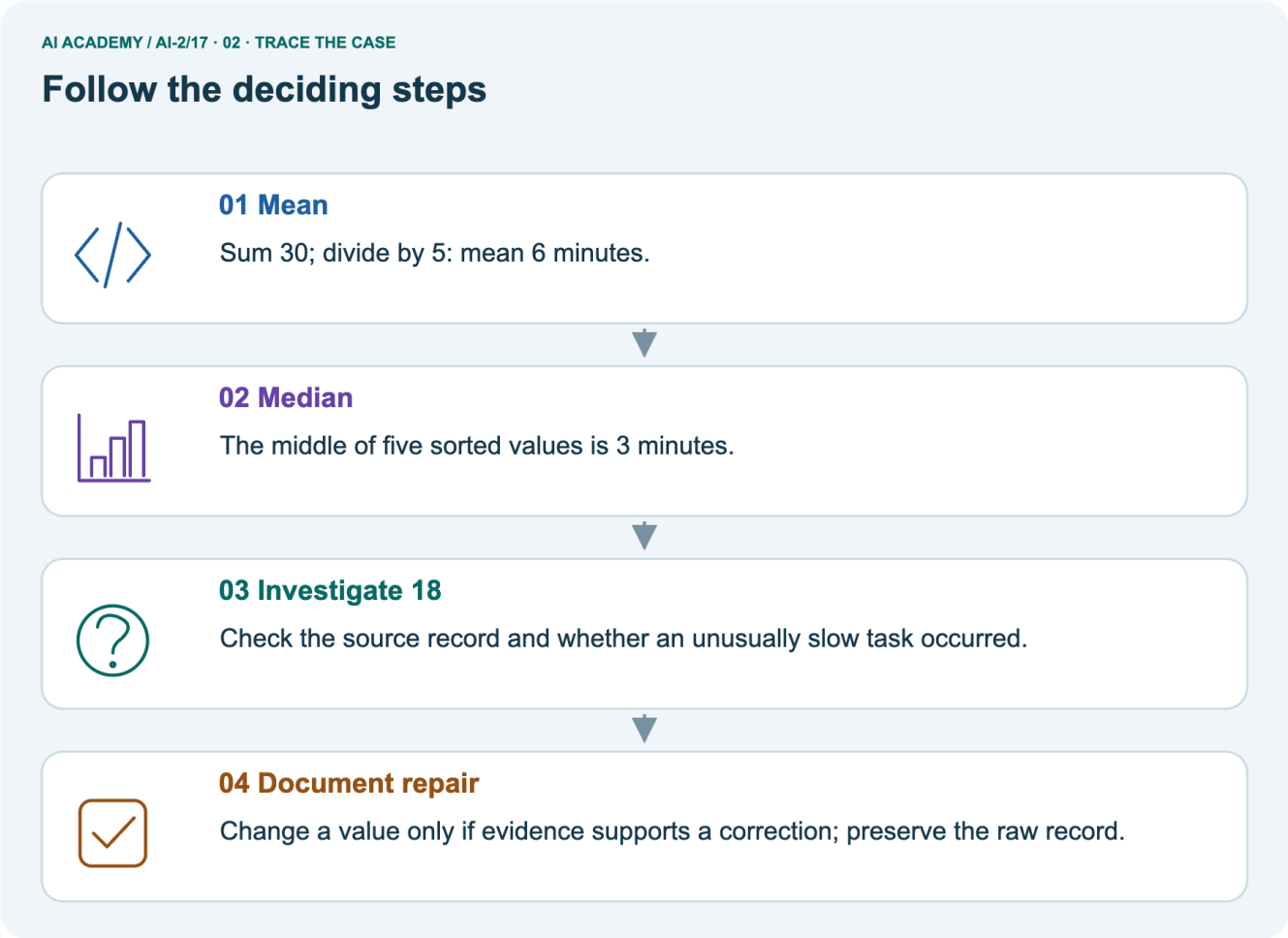
A fictional library records five book-**process**ing times in minutes. Supplied facts: Sorted times: 2,3,3,4,18. All are recorded measurements; no **evidence** yet shows 18 is a mistake.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

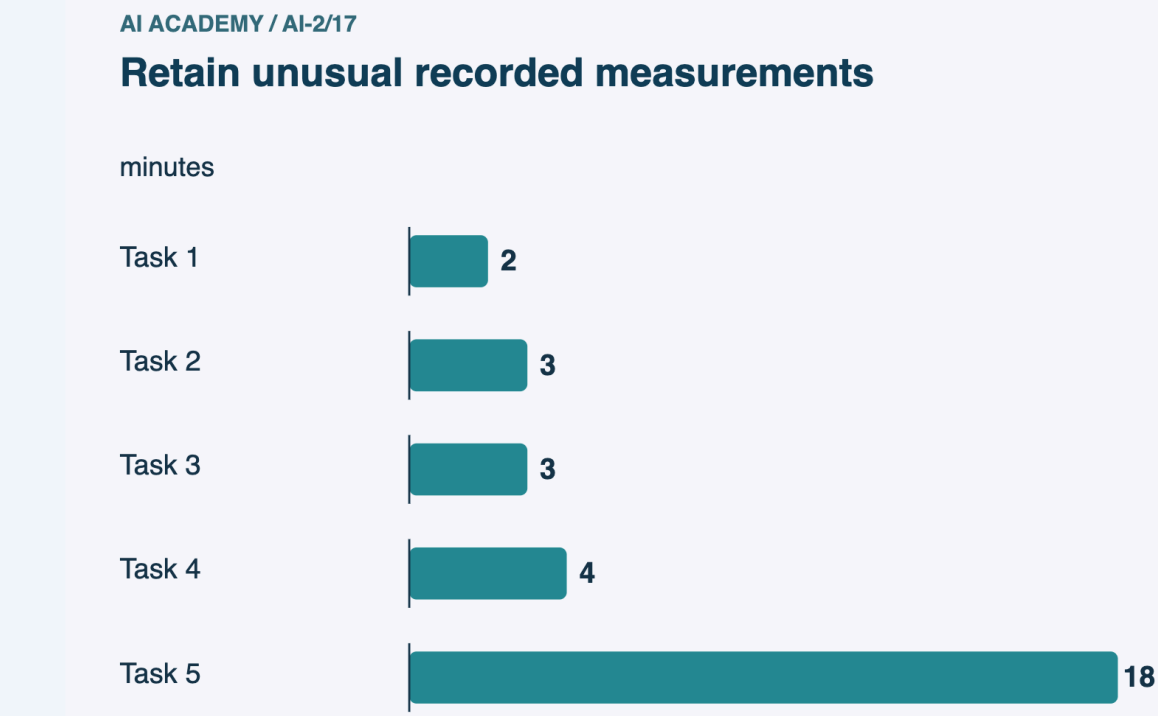
Mean: Sum 30; divide by 5: **mean** 6 minutes. → **Median**: The middle of five sorted values is 3 minutes. → **Investigate** 18: Check the source record and whether an unusually slow task occurred. → Document repair: Change a value only if **evidence** supports a correction; preserve the raw record.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Retain unusual recorded measurements



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain unusual recorded measurements. Units: minutes. Task 1: 2. Task 2: 3. Task 3: 3. Task 4: 4. Task 5: 18.

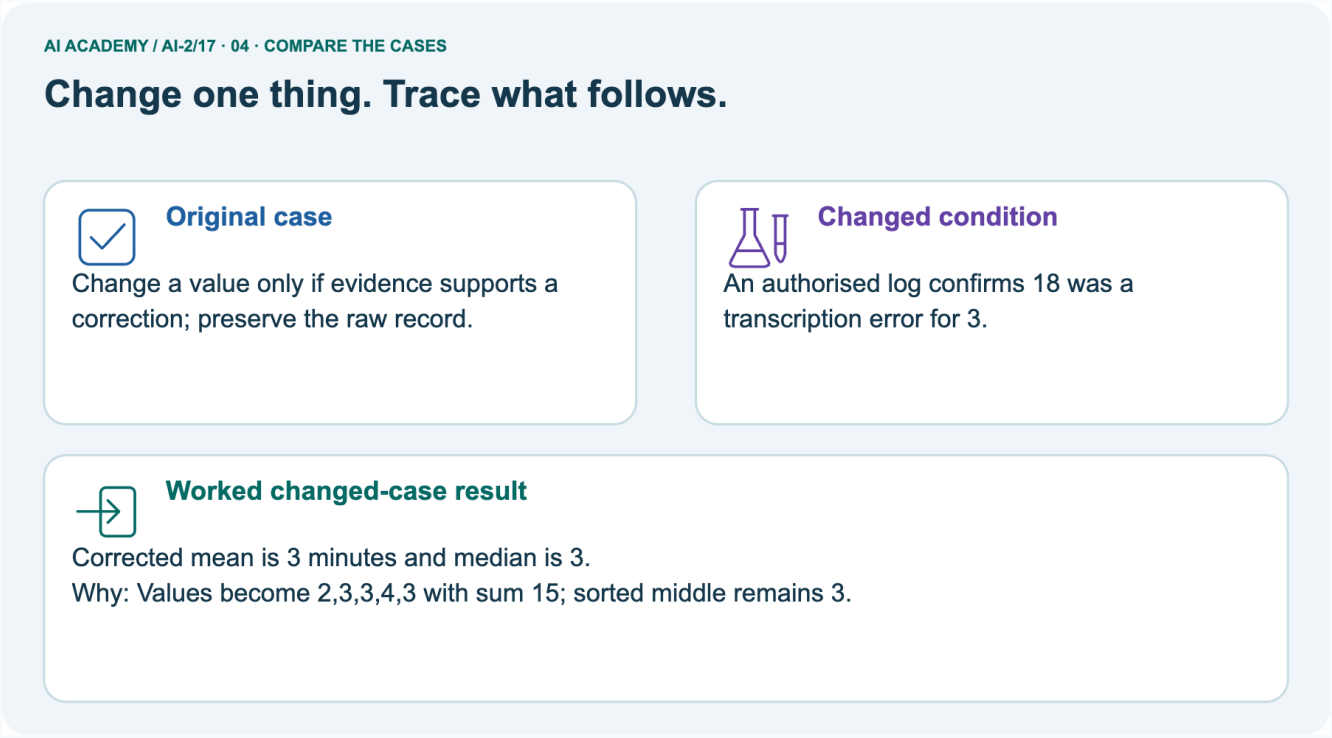
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column **mean**ings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Change a value only if **evidence** supports a correction; preserve the raw record.
Changed condition: An authorised log confirms 18 was a transcription error for 3.

Expected result: Corrected **mean** is 3 minutes and **median** is 3. Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

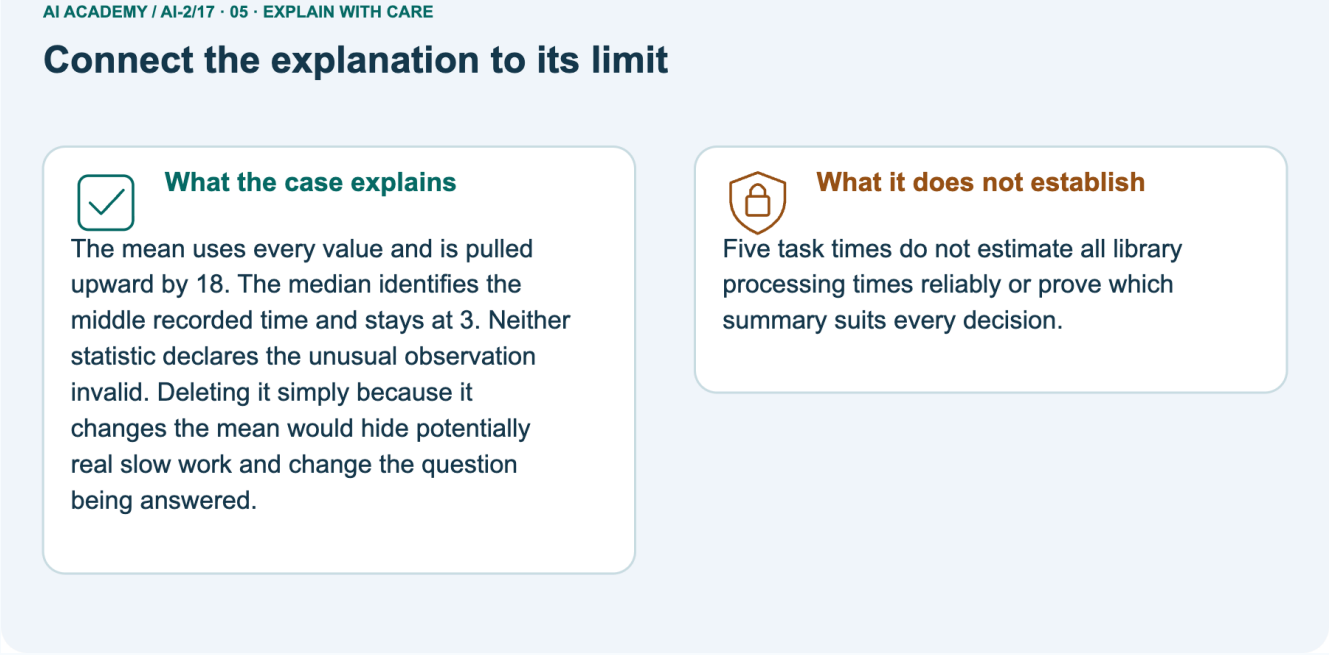
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Limit: Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

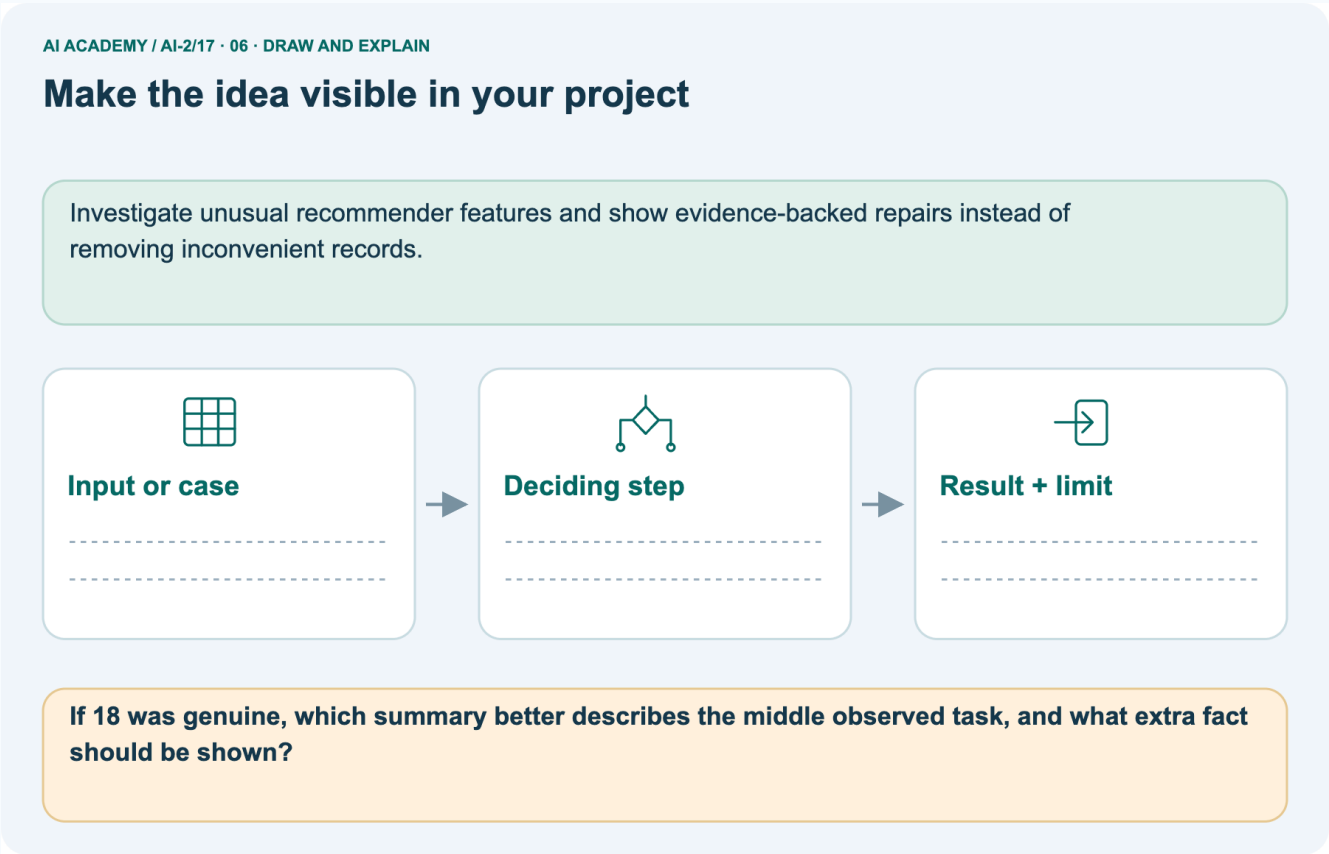
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records. Draw your **input**, deciding step, result and limit.

Supported question: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: **Median** 3 describes the middle; also report the unusual 18-minute task and the **mean** 6 rather than concealing spread.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.